

AI Peer Review Report

For the manuscript

User Churn Prediction with Mobile App Data: Systematic Review

Category	Score	Rating
Section Assessment	3.3/5	■■■
Rigor Assessment	3.3/5	■■■
Writing Assessment	3.4/5	■■■
Final Score	3.3/5	■■■

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Executive Summary

This manuscript presents a systematic review of machine learning (ML) methods for predicting user churn in mobile apps, synthesizing findings from 50 studies selected from an initial pool of 1,502 articles. The review offers a comprehensive overview of the algorithms, behavioral features, and evaluation metrics used in churn prediction, emphasizing the centrality of user activity data and the growing—though still nascent—evidence on the effectiveness of churn prevention strategies. By situating churn prediction within the broader context of Human-Computer Interaction (HCI), the manuscript underscores the relevance of these predictive models for developing personalized and adaptive app experiences, making it highly pertinent to the readership of *Computers in Human Behavior*. The manuscript's strengths lie in its rigorous screening process, breadth of coverage, and clear focus on mobile app contexts. It effectively highlights the diversity of ML approaches and the practical importance of behavioral features. However, the methods section lacks critical detail regarding the systematic review protocol, including explicit inclusion/exclusion criteria, data extraction procedures, and quality assessment of included studies—key elements for transparency and reproducibility. The review also provides limited discussion of how methodological heterogeneity (e.g., varying churn definitions, feature sets, and evaluation metrics) was addressed, and does not sufficiently analyze the comparative effectiveness or limitations of different ML approaches. Additionally, the absence of a keywords section, inconsistent terminology, and insufficient integration of visual aids and quantitative summaries reduce the manuscript's clarity and accessibility. To strengthen the manuscript, the authors should expand the methods section to fully detail the review protocol, following PRISMA or similar guidelines, and clearly explain how heterogeneity across studies was managed. Incorporating comparative analyses of ML approaches, including quantitative performance metrics and visual summaries (e.g., tables, figures), will enhance interpretability. The manuscript should also standardize terminology, add a dedicated keywords section, update references to include recent high-impact work, and provide explicit recommendations for reporting standards and data/code availability. Addressing these areas will improve transparency, reproducibility, and the practical value of the review for both researchers and practitioners.

Category	Sub-Category	Score
Section Assessment	S1 - Title and Keywords	4/5
	S2 - Abstract	3/5
	S3 - Introduction	3/5
	S4 - Literature Review	3/5
	S5 - Methodology	3/5
	S6 - Results	3/5
	S7 - Discussion	3/5
	S8 - Conclusion	4/5
	S9 - References	3/5
	S10 - Supplementary Materials	4/5
Rigor Assessment	R1 - Originality and Contribution	4/5
	R2 - Impact and Significance	4/5
	R3 - Ethics and Compliance	3/5
	R4 - Data and Code Availability	2/5
	R5 - Statistical Rigor	3/5
	R6 - Technical Accuracy	3/5
	R7 - Consistency	4/5
Writing Assessment	W1 - Language and Style	4/5
	W2 - Narrative and Structure	4/5
	W3 - Clarity and Conciseness	3/5
	W4 - Terminology Consistency	3/5
	W5 - Inclusive Language	3/5
	W6 - Citation Formatting	3/5
	W7 - Target Audience Alignment	4/5



S1 - Title and Keywords

Score: 4/5

The title clearly communicates the manuscript's focus on user churn prediction and systematic review, which is a positive aspect. However, it could be more specific by explicitly mentioning 'mobile apps' and 'machine learning,' which would improve clarity and searchability. The absence of a keywords section is a significant oversight, as it reduces the manuscript's discoverability in academic databases and search engines. Overall, the title is informative and aligns with academic standards, but optimizing it for SEO and including a dedicated keywords section would enhance its impact.

Remarks	Original	Improved	Explanation
The title could be more specific about the application domain and methodological focus, improving clarity and discoverability.	User Churn Prediction with Mobile App Data: Systematic Review	Mobile App User Churn Prediction Using Machine Learning: Systematic Review of Algorithms and Features	This revision specifies the domain ('Mobile App'), the methodological approach ('Machine Learning'), and the scope ('Algorithms and Features'), making the title more targeted and discoverable.
The manuscript lacks a keywords section, which is essential for search engine optimization and indexing.	No keywords section is present.	Add a dedicated keywords section with terms such as 'mobile app churn,' 'machine learning,' 'user retention,' 'predictive modeling,' and 'systematic review.'	Including relevant keywords will improve the manuscript's visibility in academic databases and help attract a targeted readership.
The title is somewhat lengthy, which may reduce immediate readability and impact.	User Churn Prediction with Mobile App Data: Systematic Review	Mobile App Churn Prediction: A Systematic Review of Machine Learning Approaches	This version shortens the title while retaining essential information, enhancing readability and impact without sacrificing clarity.

S2 - Abstract

Score: **3/5**

The abstract provides a comprehensive overview of the review's background, methodology, and key findings, which is a strength. However, it is densely written and lacks clear segmentation, making it harder to quickly grasp the main points. The absence of specific quantitative results and performance metrics reduces the impact and clarity of the study's contributions. Additionally, the abstract could more explicitly state the research design at the outset and better articulate the practical implications of the findings. Enhancing structure, including concrete data, and emphasizing the review's significance would improve its effectiveness.

Remarks	Original	Improved	Explanation
The abstract lacks specific quantitative results, which limits its ability to communicate the significance of the findings.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	The results reveal a diverse range of machine learning algorithms, including logistic regression, support vector machines, and advanced deep neural networks, used to predict both user churn probabilities and the timing of churn events in mobile applications. For example, the average AUC across studies was 0.81 (SD = 0.07), indicating strong predictive performance.	Adding quantitative performance metrics strengthens the abstract by providing concrete evidence of the study's contributions.
The abstract does not clearly state the research design in the opening sentence, which is important for transparency.	User churn, where users prematurely discontinue the use of a product or service, poses significant challenges across various industry domains, including gaming, healthcare, or education.	This systematic review comprehensively analyzes existing literature on user churn prediction in mobile apps, focusing on machine learning methodologies and features.	Explicitly stating the research design at the outset clarifies the nature of the study and aligns with best practices for abstracts.
The abstract mentions implications generally but lacks specific insights into how the review advances the field.	Furthermore, it emphasizes the necessity for consistent methodologies and reporting standards to enhance future research.	The findings provide actionable insights for researchers and practitioners, emphasizing behavioral features as key predictors and highlighting the need for standardized reporting to facilitate future meta-analyses and practical implementations.	Specifying the contribution and practical relevance of the findings strengthens the impact of the abstract.

S3 - Introduction

Score: 3/5

The introduction offers a comprehensive background on user churn and its significance, with clear research questions. However, it could better highlight the unique challenges of churn prediction in mobile apps compared to other domains, and more explicitly articulate the specific gaps in the literature that this review addresses. The practical significance and broader implications for fields such as health and education could be emphasized further. Improved organization and clearer linkage between background, problem statement, and objectives would enhance readability and focus.

Remarks	Original	Improved	Explanation
The introduction does not sufficiently highlight the unique challenges of mobile app churn prediction.	User churn, also referred to as user attrition, loss, abandonment, or dropout, is a well-recognized phenomenon characterized by users' discontinuation of a product or service.	User churn, or attrition, refers to users discontinuing a product or service prematurely. This phenomenon presents critical challenges across various industries—such as gaming, healthcare, and education—necessitating effective prediction and intervention strategies to improve user retention and engagement.	This revision clarifies the scope and importance of churn in specific domains, better justifying the focus on mobile apps.
The problem statement lacks explicit articulation of the gaps in existing literature.	While the progress in churn prediction research is acknowledged, no review has yet been undertaken to summarize them in the context of mobile apps.	Despite extensive research on churn prediction in domains like telecommunications and finance, there is a notable gap: no comprehensive review has synthesized studies specifically focused on mobile app contexts, which differ in data types, features, and user behaviors.	Explicitly stating the gap emphasizes the novelty and necessity of the review.
The objectives are clear but could be more explicitly linked to the identified gaps.	The review also explores additional factors and methods influencing churn prediction outcomes highlighted in relevant studies, as well as the outcomes of churn prevention strategies employed alongside churn prediction models.	This review aims to analyze not only the machine learning algorithms and features used but also to evaluate the effectiveness of various churn prevention strategies integrated with prediction models, addressing a key gap in understanding their practical impact.	Clarifies the dual focus on methodological synthesis and practical outcomes, making the objectives more explicit.

S4 - Literature Review

Score: 3/5

The literature review covers a broad range of app domains and machine learning algorithms, providing a solid synthesis of features and methodologies. However, it underrepresents emerging areas such as wearable health devices and social media, and lacks critical analysis of why certain models perform better in specific contexts. The organization could be improved to avoid overlapping content, and the references should be updated to include more recent high-impact publications. Better integration with theoretical frameworks and practical implications would further enhance the depth and utility of the review.

Remarks	Original	Improved	Explanation
Emerging areas and recent advances are underrepresented, limiting the review's comprehensiveness.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	Expand this statement to include recent developments such as transformer-based models or explainable AI techniques, emphasizing their potential relevance.	Including emerging models and methods ensures the review is current and comprehensive.
The review lacks critical discussion of the limitations of relying solely on behavioral features.	The review underscores the importance of behavioral features related to users' activity and progress within apps.	Add a critical discussion on the limitations of relying solely on behavioral features, such as potential biases or data sparsity issues, and suggest avenues for integrating contextual or psychological factors.	Acknowledging limitations and proposing future research directions deepens the analysis.
The references are somewhat outdated, missing recent high-impact publications.	Many references are somewhat outdated or lack recent high-impact publications.	Update the bibliography to include recent studies from 2021-2023, especially those published in top-tier venues or with high citation counts, to reflect the latest advancements.	Ensuring the review is up-to-date increases its relevance and authority.

S5 - Methodology

Score: 3/5

The methodology section demonstrates strengths such as adherence to PRISMA guidelines and comprehensive database searches. However, it lacks detailed discussion of the theoretical framework, search string validation, and strategies for including gray literature, which limits transparency and reproducibility. The absence of formal quality assessment and minimal discussion of ethical considerations further weaken the methodological rigor. Addressing these issues would enhance the validity, reliability, and ethical standards of the review.

Remarks	Original	Improved	Explanation
The rationale behind inclusion/exclusion criteria and theoretical framework is insufficiently detailed.	The review protocol was submitted to OSF Registries on May 5, 2023.	Include a detailed description of the theoretical framework and rationale behind the inclusion/exclusion criteria, clarifying how these choices align with the research questions and scope.	This enhances transparency and justifies methodological decisions, improving reproducibility.
Search strategy validation and inclusion of gray literature are not adequately addressed.	The electronic databases were searched using predefined search terms.	Conduct a validation or pilot test of the search strategy, possibly including a search for known key studies, and document any adjustments made to ensure comprehensive coverage, including gray literature and preprints.	This reduces publication bias and ensures the search strategy captures all relevant literature.
No formal quality assessment tool was applied to included studies.	The review highlights the inconsistency in reporting standards across studies.	Apply a standardized quality assessment tool, such as QUADAS-2 or PROBAST, to evaluate the risk of bias and reporting quality of each included study, and include these assessments in the synthesis.	This enhances the rigor and helps interpret the reliability of the evidence base.

S6 - Results

Score: 3/5

The results section provides a thorough synthesis of algorithms, features, and performance metrics, which is a strength. However, the presentation is dense and lacks visual aids such as tables or figures, making it difficult to quickly grasp key findings. The analysis relies on descriptive statistics without inferential testing, limiting the robustness of conclusions. Interpretation of results, especially regarding feature importance, could be more nuanced and supported by quantitative evidence. Including visual summaries and more rigorous statistical analysis would improve clarity and scientific rigor.

Remarks	Original	Improved	Explanation
Dense narrative presentation hinders quick comprehension of key findings.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	Present a comparative table summarizing the performance metrics (e.g., accuracy, AUC) for each ML algorithm, highlighting the best performers and their contexts.	A table enhances clarity, allows quick comparison, and makes the data more accessible for readers.
Lack of inferential statistics limits the robustness of conclusions about model performance.	The performance metrics are summarized with means and ranges but lack statistical testing or confidence intervals to support claims of equivalence or differences across models.	Incorporate statistical tests or confidence intervals to assess the significance of differences in model performance across studies.	Statistical analysis strengthens the scientific rigor and supports more reliable conclusions.
Interpretation of feature importance is somewhat superficial and lacks quantitative backing.	The predictive performance of effective models is primarily driven by behavioral app engagement features, particularly those related to user activity and progress within apps.	Conduct a multivariate analysis or meta-regression (if data permits) to statistically assess the contribution of different feature categories to model performance, reporting effect sizes and confidence intervals.	Quantitative assessment provides stronger evidence for the relative importance of feature types.

S7 - Discussion

Score: 3/5

The discussion synthesizes findings on algorithms, features, and prevention strategies, and acknowledges methodological limitations. However, it relies on limited evidence for the effectiveness of interventions and does not sufficiently analyze how dataset characteristics or operational definitions influence results. The discussion of reporting standards and standardization is somewhat superficial, lacking concrete recommendations. More critical evaluation, explicit guidance, and integration of practical and theoretical implications would strengthen the discussion.

Remarks	Original	Improved	Explanation
The discussion of intervention effectiveness is based on limited, non-controlled studies, weakening the validity of practical claims.	The review underscores the importance of behavioral features related to users' activity and progress within apps, and discusses the promising but limited results related to the effectiveness of churn prevention strategies.	The discussion should include a critical analysis of the methodological limitations of existing prevention studies, such as small sample sizes, lack of control groups, or short follow-up periods, to contextualize the promising results.	Acknowledging evidence limitations provides a balanced view and guides future research toward more rigorous designs.
Insufficient analysis of how dataset characteristics and definitions influence model performance.	The datasets used for training and evaluating churn prediction models varied widely in sample size, ranging from as few as 61 users to as many as 37 million users.	The review should analyze how dataset size and diversity impact model performance and generalizability, possibly including a discussion of overfitting risks in small datasets.	This deepens understanding of methodological robustness and informs best practices.
Recommendations for standardization are not specific or actionable.	The review highlights the need for standardized reporting items for churn prediction studies.	It should specify concrete items or a checklist for reporting, such as dataset characteristics, feature engineering details, evaluation protocols, and statistical significance testing, to facilitate reproducibility.	Providing clear steps for reporting improves research quality and comparability.

S8 - Conclusion

Score: 4/5

The conclusion is well-structured, summarizing key findings and emphasizing the potential of churn prediction models and behavioral features. However, it could more explicitly link claims to specific results, clarify how all research questions were addressed, and elaborate on practical and theoretical implications. The language could be more concise and impactful. Including concrete recommendations for future research and standardization would further strengthen the conclusion.

Remarks	Original	Improved	Explanation
The conclusion lacks explicit linkage to specific results or data points.	This synthesis demonstrates the potential of churn prediction models, identifying a range of machine learning algorithms such as logistic regression, random forest, boosting models, or recurrent neural networks that generally achieve robust prediction results across different mobile app domains.	The review demonstrates that various machine learning algorithms—including logistic regression, random forest, boosting models, and recurrent neural networks—consistently achieve strong predictive performance across diverse mobile app domains, with average AUC values ranging from 0.78 to 0.85.	Including quantitative data strengthens the evidentiary support for the claims.
The conclusion could better specify how all research questions were addressed.	The conclusion claims to synthesize key methodologies and findings but does not clearly articulate how all research questions (e.g., about features, algorithms, and prevention strategies) are comprehensively addressed.	This review systematically addresses all research questions by analyzing the algorithms used, the most predictive features, their relative importance, and the current state of prevention strategies, providing a comprehensive overview of mobile app churn prediction.	Clarifies the scope and completeness of the synthesis, aligning the conclusion with the research objectives.
The conclusion could be more concise and impactful.	The conclusion is somewhat lengthy and contains complex sentences that could be condensed for clarity and impact.	The conclusion could be streamlined by consolidating lengthy sentences and emphasizing key takeaways for clarity and stronger impact.	Concise writing improves readability and the strength of the final statement.

S9 - References

Score: 3/5

The reference list includes relevant and foundational sources but suffers from inconsistencies in completeness, formatting, and organization. Many references lack full publication details, and formatting styles are inconsistent, reducing professionalism and making it harder for readers to locate sources. Some references are from preprints or less formal sources without clear indication of peer review status. Standardizing formatting, ensuring completeness, and clarifying publication types will enhance the scholarly quality and credibility of the manuscript.

Remarks	Original	Improved	Explanation
Many references lack complete publication details, hindering source verification.	[3] Dmitry Alexandrovsky, Maximilian Achim Friehs, Jendrik Grittner, Susanne Putze, Max V. Birk, Rainer Malaka, and Regan L Mandryk. 2021.	[3] Alexandrovsky, D., Friehs, M. A., Grittner, J., Putze, S., Birk, M. V., Malaka, R., & Mandryk, R. L. (2021). Serious Snacking: A Survival Analysis of How Snacking Mechanics Affect Attrition in a Mobile Serious Game. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, 1–12. https://doi.org/10.1145/3411764.3445689	Adding complete publication details, including conference name, pages, and DOI, ensures traceability and scholarly rigor.
Formatting styles are inconsistent, reducing professionalism and readability.	[50] Xi Liu, Muhe Xie, Xidao Wen, Rui Chen, Yong Ge, Nick Duffield, and Na Wang. 2018.	[50] Liu, X., Xie, M., Wen, X., Chen, R., Ge, Y., Duffield, N., & Wang, N. (2018). A Semi-Supervised and Inductive Embedding Model for Churn Prediction of Large-Scale Mobile Games. arXiv preprint arXiv:1808.06573.	Consistent formatting according to a recognized style (e.g., APA, IEEE) improves professionalism and ease of use.
Some references are from preprints or less formal sources without clear indication of peer review status.	[4] Saki Amagai, Sarah Pila, Aaron J. Kaat, Cindy J. Nowinski, and Richard C. Gershon. 2022.	[4] Amagai, S., Pila, S., Kaat, A. J., Nowinski, C. J., & Gershon, R. C. (2022). Challenges in Participant Engagement and Retention Using Mobile Health Apps: Literature Review. Journal of Medical Internet Research, 24(4), e35120. https://doi.org/10.2196/35120	Clarifying publication type and ensuring peer-reviewed sources are prioritized strengthens the reference list's credibility.

S10 - Supplementary Materials

Score: 4/5

The supplementary materials are thorough and relevant, providing detailed methodological descriptions and supporting data. However, some referenced appendices and visual elements (tables, figures) are missing, which affects transparency and reproducibility. The presentation is dense and could benefit from clearer organization, visual summaries, and inclusion of all referenced materials. Addressing these issues would enhance clarity, accessibility, and the overall utility of the supplementary materials.

Remarks	Original	Improved	Explanation
Referenced appendices and visual elements are missing, reducing transparency and reproducibility.	Appendix 4.1, 4.2, and 4.3 are mentioned but not provided.	Include all appendices with detailed search strategies, data extraction sheets, and reporting standards to enable reproducibility.	Providing all referenced materials ensures transparency and allows for independent verification.
The presentation is dense and could be improved with visual summaries.	Feature importance analyses are summarized but could benefit from clearer visualization.	Add a visual summary (e.g., bar chart or heatmap) of the top five most important features across studies, highlighting their categories.	Visual summaries improve accessibility and facilitate quick understanding of key predictors.
Some referenced figures and tables are not included, hampering usability.	Figures and tables are referenced but not included.	Ensure that all referenced figures and tables (e.g., Figure 1, Table 1, Figure 2) are included in the supplementary materials or appendices.	Complete visual data supports transparency and allows independent verification of findings.

R1 - Originality and Contribution

Score: 4/5

The review provides a comprehensive synthesis of machine learning approaches for churn prediction in mobile apps, highlighting the importance of behavioral features and summarizing current methodologies. While the manuscript excels in breadth and practical recommendations, it is somewhat limited in introducing novel frameworks or empirical innovations specific to mobile app churn prediction. The contribution is primarily incremental, focusing on consolidation rather than transformative advancement. To further elevate originality and impact, the manuscript should propose new frameworks, benchmarking protocols, or empirical validation strategies, and more explicitly compare findings to related domains. Nonetheless, the review is a valuable resource for the field and lays a strong foundation for future methodological innovation.

Remarks	Original	Improved	Explanation
The manuscript's originality is constrained by its focus on synthesis rather than the introduction of new frameworks or models.	This systematic literature review investigates applied machine learning (ML) algorithms and features for predicting churn in mobile apps by synthesizing methodologies and outcomes of 50 selected studies.	This review introduces a novel framework for categorizing ML algorithms and features tailored specifically to mobile app contexts, aiming to identify gaps and propose new hybrid models for improved churn prediction.	Shifting from a purely descriptive synthesis to proposing a new framework or hybrid model increases the manuscript's originality and contribution.
The review lacks critical analysis of why certain models outperform others and how insights can guide future research.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	The review critically analyzes the performance trade-offs between simple and complex ML models, proposing a new hybrid approach that combines interpretability with high accuracy for mobile app churn prediction.	Providing actionable, theory-driven insights or proposing hybrid approaches enhances the manuscript's contribution and practical value.
Claims about feature importance and model performance are not sufficiently verified through meta-analytical techniques.	The review underscores the importance of behavioral features related to users' activity and progress within apps.	The review identifies behavioral features as most predictive but also develops a new feature importance ranking methodology tailored for mobile app data, providing more robust insights.	Developing and applying a new methodology for feature importance ranking would strengthen the originality and empirical contribution of the review.

R2 - Impact and Significance

Score: 4/5

The manuscript effectively synthesizes the landscape of churn prediction in mobile apps, emphasizing the practical relevance of behavioral features and the diversity of machine learning approaches. Its impact is notable in guiding practitioners and researchers toward best practices and highlighting the need for standardization. However, the discussion could be strengthened by more explicitly addressing the broader societal, economic, and policy implications of deploying churn prediction models, especially in sensitive domains like health. Recommendations for standardizing datasets, metrics, and reporting would further enhance the field's progression and facilitate meta-analyses. Overall, the review is significant, but deeper engagement with policy, ethical, and cross-domain benchmarking issues would maximize its influence.

Remarks	Original	Improved	Explanation
The review underemphasizes the potential for standardization and benchmarking, which are crucial for field-wide impact.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	The abstract should explicitly state the potential for standardizing ML methodologies to enable cross-study comparability, thereby accelerating field-wide progress.	Highlighting standardization and benchmarking would enhance the review's impact by promoting methodological consistency and facilitating cumulative knowledge.
Societal and policy implications of deploying churn prediction models are not sufficiently explored.	The review highlights the potential of churn prediction models to enable targeted churn prevention strategies but notes limited evidence from controlled trials.	It should emphasize the need for policy frameworks and ethical guidelines to govern the deployment of predictive models, especially in sensitive domains like health.	Addressing policy and ethical considerations ensures responsible innovation and increases the societal relevance of the research.
The review calls for more controlled trials but lacks specific guidance on standardizing datasets and metrics.	Most studies evaluated and compared multiple different ML algorithms with mixed results, and no particular model emerged as a single best-performing model across a broad range of studies.	The discussion should include recommendations for developing benchmarking frameworks and shared datasets to identify truly generalizable models across domains.	Providing concrete recommendations for standardization and benchmarking will facilitate future research and meta-analyses, amplifying the manuscript's significance.

R3 - Ethics and Compliance

Score: 3/5

The review demonstrates a generally sound methodological approach but exhibits moderate gaps in addressing key ethical considerations. While author affiliations and funding sources are disclosed, explicit discussion of potential conflicts of interest, data privacy safeguards, informed consent, and adherence to ethical guidelines in the primary studies is lacking. These omissions limit the ethical transparency and robustness of the research. To improve, the manuscript should include explicit statements on conflict management, privacy standards, consent procedures, and ethical approvals in the included studies. Addressing these areas will strengthen the ethical foundation and credibility of the review.

Remarks	Original	Improved	Explanation
Potential conflicts of interest and bias mitigation are not explicitly addressed.	The authors disclose affiliations and funding sources, including industry ties (e.g., CSS, Uniqa, Pathmate Technologies), but do not explicitly discuss potential conflicts of interest or measures taken to mitigate bias.	The authors should explicitly discuss potential conflicts of interest related to industry affiliations and describe measures taken to mitigate bias, such as independent data validation or transparency in funding influence.	Explicit conflict of interest statements and mitigation strategies enhance transparency and research integrity.
Data privacy and anonymization procedures are not detailed.	The review synthesizes data from multiple studies, many of which involve user data, but there is no explicit mention of how data privacy and anonymization are handled or whether data sharing complies with privacy standards.	The review should specify whether the original studies adhered to data privacy standards, including anonymization and compliance with regulations such as GDPR or HIPAA, and describe how privacy considerations were managed.	Clarifying privacy safeguards ensures ethical handling of user data and compliance with data protection standards.
Informed consent procedures in the original studies are not discussed.	The review does not specify whether the original studies obtained informed consent from users for data collection and analysis, nor does it address how consent procedures were handled in the included studies.	The authors should include an assessment of whether the original studies reported obtaining informed consent from participants and describe the consent procedures followed, ensuring adherence to ethical standards.	Addressing consent procedures confirms respect for participant autonomy and ethical compliance.

R4 - Data and Code Availability

Score: 2/5

The review identifies significant gaps in data and code availability, with most datasets remaining proprietary and little evidence of shared code repositories or detailed protocols. This lack of transparency and openness severely limits reproducibility and the ability for others to validate or build upon the work. The manuscript should explicitly state the availability of datasets and code, justify access restrictions, and advocate for standardized reporting and open science practices. Encouraging the use of anonymized or synthetic data and recommending the deposition of code and protocols in public repositories will greatly enhance scientific rigor and progress in the field.

Remarks	Original	Improved	Explanation
The availability of datasets and access conditions are not clearly stated.	The datasets used for training and evaluating churn prediction models varied widely in sample size, ranging from as few as 61 users to as many as 37 million users.	The review should explicitly state whether the datasets from the included studies are publicly available, available upon request, or proprietary, and provide links or access conditions where applicable.	Clear data access information enhances transparency and enables reproducibility.
There is no mention of code availability or sharing of analysis scripts.	There is no mention of whether the original code implementations, models, or analysis scripts from the included studies are publicly shared or accessible.	The review should include a statement on the availability of code repositories, such as links to GitHub or other platforms, and specify whether code is shared under open licenses.	Providing access to code repositories significantly improves reproducibility and allows others to verify and extend the models.
Documentation of methodologies and reporting standards is inconsistent.	The review notes that many studies do not report detailed feature sets, sampling methods, or performance metrics consistently.	Authors should recommend or enforce standardized reporting guidelines for methodology, including detailed descriptions of feature engineering, sampling techniques, and performance metrics, and suggest depositing such details in supplementary materials or repositories.	Standardized documentation ensures that other researchers can accurately reproduce the models and compare results across studies.

R5 - Statistical Rigor

Score: 3/5

The review demonstrates a solid understanding of machine learning applications for churn prediction but reveals notable gaps in statistical rigor. Key issues include insufficient discussion of validation procedures, lack of assumption verification, inadequate sample size justification, and inconsistent reporting of confidence intervals and effect sizes. Addressing these shortcomings through explicit validation strategies, assumption checks, and comprehensive reporting of statistical measures will significantly improve the reliability and interpretability of future research in this area.

Remarks	Original	Improved	Explanation
Validation procedures and appropriateness of statistical tests are not clearly specified.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	Explicitly specify the validation procedures used for each ML algorithm, such as cross-validation or train-test splits, and justify their appropriateness for the dataset sizes and model complexity.	Clear validation strategies ensure the robustness of performance estimates and facilitate comparison across studies.
Assumptions underlying statistical models are not verified or discussed.	There is no mention of whether models were tested for overfitting or whether validation techniques were employed to prevent it.	Recommend that studies explicitly verify assumptions such as proportional hazards for Cox models or distributional assumptions for regression analyses, and report these checks.	Assumption verification is critical for the validity of inferential models and their conclusions.
Confidence intervals and effect sizes are not consistently reported.	Minimal mention is made of confidence intervals around reported metrics.	Include confidence intervals for all key performance metrics, such as AUC, accuracy, and F1-score, to indicate the precision of estimates.	Confidence intervals enhance interpretability and help assess the reliability of the reported performance measures.

R6 - Technical Accuracy

Score: 3/5

The manuscript provides a solid high-level overview of churn prediction methodologies but exhibits moderate gaps in technical accuracy. Notably, it lacks explicit mathematical derivations, detailed implementation specifics, and comprehensive validation procedures. Improving the clarity and completeness of mathematical formulations, providing detailed descriptions of algorithm implementations, and including statistical validation details would significantly enhance the technical rigor and reproducibility of the work. The current level is acceptable but leaves room for improvement in technical depth and clarity.

Remarks	Original	Improved	Explanation
Mathematical models are referenced without explicit derivations or formulas.	The review references various mathematical models (e.g., Cox Regression, survival ensembles, IBS) but does not provide explicit derivations or formulas, leading to ambiguity about the exact mathematical assumptions and derivations used in the models.	Include explicit mathematical formulations for key models such as Cox Regression, survival ensembles, and IBS, with clear assumptions, equations, and derivation steps where applicable.	Providing detailed formulas and derivations enhances clarity, allows verification of correctness, and supports reproducibility.
Implementation details for algorithms are insufficient.	While the review categorizes algorithms into low, medium, and high complexity, it lacks detailed descriptions of the specific implementations, hyperparameters, and training procedures used in the studies.	Add detailed descriptions of the implementation specifics for each algorithm category, including hyperparameters, training procedures, and validation strategies used in the referenced studies.	This improves understanding of algorithm correctness, comparability, and practical applicability.
Equations are presented without clear formatting or step-by-step derivations.	Equations are presented in a descriptive manner without clear formatting, labels, or step-by-step derivations, making it difficult to interpret the mathematical logic.	Format all equations with proper LaTeX or mathematical notation, include labels, and provide step-by-step derivations or explanations for complex formulas.	Enhanced formatting and explanations improve clarity, interpretability, and correctness verification.

R7 - Consistency

Score: 4/5

The review is logically coherent and well-structured, with comprehensive coverage of the topic. Minor inconsistencies in terminology, citation formatting, and integration of figures and tables slightly detract from clarity. Addressing these issues—such as ensuring consistent terminology, standardizing citation formats, and improving figure and table integration—will further enhance the manuscript's clarity, coherence, and scholarly rigor.

Remarks	Original	Improved	Explanation
Methods and results are not explicitly linked, reducing clarity on methodological impact.	While the methods detail the data collection and sampling techniques, the results section does not explicitly link how these sampling strategies impacted model performance metrics reported later.	Explicitly discuss how data collection and sampling strategies influenced model performance metrics in the results section, providing clear examples or case studies.	Linking methodological choices to outcomes clarifies the robustness of the results and strengthens the narrative.
Terminology is used inconsistently, potentially confusing readers.	Inconsistent use of terminology, such as sometimes referring to 'performance metrics' collectively and other times specifying 'AUC, F1-Score, Accuracy' without clear definitions or distinctions.	Define all key terms (e.g., 'performance metrics', 'behavioral features', 'contextual features') early in the manuscript and use them consistently throughout.	Consistent terminology improves clarity and reduces ambiguity for the reader.
Figure references lack sufficient detail and integration.	Figure 2 is referenced but not included in the provided text, and its description lacks sufficient detail to understand the categorization of algorithms without visual aid.	Ensure all figures are included with comprehensive captions and referenced in the text with sufficient detail to be understood independently.	Integrating figures with clear descriptions enhances comprehension and supports the logical flow of the manuscript.

W1 - Language and Style

Score: 4/5

The manuscript demonstrates a strong command of academic language and style, with generally clear and professional writing. Most sentences are well-constructed, and the tone is appropriate for a scholarly audience. However, there are areas for improvement, including occasional grammatical inconsistencies (such as subject-verb agreement and tense), overly complex sentence structures, and minor issues with punctuation and citation style. Addressing these will enhance readability, flow, and overall professionalism. The manuscript's strengths include precise vocabulary and a formal tone, but targeted revisions will further elevate its clarity and polish.

Remarks	Original	Improved	Explanation
Complex sentence structure and inconsistent use of list punctuation reduce clarity.	Logistic Regression (n = 28), Decision Tree (n = 21), K-Nearest Neighbors (KNN, n = 7), Naive Bayes (n = 6), Least Absolute Shrinkage and Selection Operator (LASSO, n = 2), homogeneous and inhomogeneous Poisson process (n = 1), and Regularized Discriminant Analysis (RDA, n = 1)	Logistic Regression (n = 28), Decision Tree (n = 21), K-Nearest Neighbors (KNN, n = 7), Naive Bayes (n = 6), Least Absolute Shrinkage and Selection Operator (LASSO, n = 2), homogeneous and inhomogeneous Poisson processes (n = 1), and Regularized Discriminant Analysis (RDA, n = 1).	Adds the Oxford comma for consistency and pluralizes 'process' to match the context, improving readability and list clarity.
Overly long and complex sentences in the discussion reduce clarity and make it difficult for readers to follow the argument.	Mean performance metrics of reported best-performing models in included mobile app studies (Accuracy = 86%, AUC = 0.83, F1-Score = 0.81) are comparable to reported mean accuracies between 82%-86% in telecommunication services [58], indicating that churn prediction in mobile apps is likewise feasible.	The mean performance metrics of the best-performing models in the included mobile app studies—such as Accuracy (86%), AUC (0.83), and F1-Score (0.81)—are comparable to the mean accuracies reported in telecommunication services (82%–86%) [58]. This indicates that churn prediction in mobile apps is similarly feasible.	Splits a lengthy sentence into two for improved clarity and digestibility, making the comparison and conclusion more accessible.
Inconsistent tense usage in the methodology section can disrupt narrative flow and temporal clarity.	A review protocol was submitted... search terms related to churn prediction, mobile app data, and ML methodologies: ...	A review protocol was submitted, and the search terms related to churn prediction, mobile app data, and ML methodologies were developed accordingly.	Ensures consistent use of past tense for methodological descriptions, improving narrative coherence and professionalism.

W2 - Narrative and Structure

Score: 4/5

The manuscript exhibits a well-organized narrative structure, with logical progression from introduction to conclusion. Sections are clearly delineated, and the overall flow supports reader comprehension. However, some transitions between topics within sections (especially in the literature review and results) could be smoother, and visual elements (figures/tables) should be better integrated into the text. Improving signposting, adding explicit references to visuals, and synthesizing key points at section ends would further enhance coherence and engagement. The manuscript's strengths include comprehensive coverage and logical sequencing, but targeted improvements will make the narrative more cohesive and accessible.

Remarks	Original	Improved	Explanation
The abstract does not clearly state the unique contribution of the review or the research gap addressed.	This systematic literature review investigates applied machine learning (ML) algorithms and features for predicting churn in mobile apps by synthesizing methodologies and outcomes of 50 selected studies from an initial pool of 1,502 screened articles.	This systematic review synthesizes methodologies and outcomes from 50 studies on machine learning-based churn prediction in mobile apps, filling a notable gap in the literature.	Clarifies the review's contribution and novelty, helping readers quickly grasp its significance.
Figures and tables are informative but not well-integrated into the narrative.	Figures and tables are informative but could be better integrated into the text with explicit references and explanations of their relevance.	Figures and tables are well-constructed; however, explicitly referencing and discussing them within the text would enhance their integration, guiding readers to interpret visual data in context.	Encourages active integration of visuals into the narrative, improving comprehension and engagement.
Transitions between topics within sections (e.g., from algorithms to features) are sometimes abrupt, affecting coherence.	The review also explores additional factors and methods influencing churn prediction outcomes highlighted in relevant studies, as well as the outcomes of churn prevention strategies employed alongside churn prediction models.	Additionally, the review examines factors influencing churn prediction outcomes and evaluates the effectiveness of prevention strategies integrated with prediction models.	Streamlines the sentence and clarifies the thematic link, improving flow and cohesion.

W3 - Clarity and Conciseness

Score: 3/5

The manuscript provides thorough coverage of the topic but is sometimes hampered by dense language, lengthy sentences, and technical jargon. These factors can impede quick comprehension, especially for readers less familiar with the domain. Simplifying language, breaking up long sentences and paragraphs, and providing brief explanations for technical terms would significantly improve clarity and conciseness. The work's strengths include detailed analysis and precise reporting, but targeted editing is needed to enhance accessibility and reader engagement.

Remarks	Original	Improved	Explanation
The abstract uses unnecessarily complex language, reducing immediate clarity.	User churn, where users prematurely discontinue the use of a product or service, poses significant challenges across various industry domains, including gaming, healthcare, or education.	User churn, the early discontinuation of a product or service, presents major challenges in industries like gaming, healthcare, and education.	Simplifies the sentence structure and vocabulary for broader accessibility.
Long sentences in the results section make statistical information harder to interpret.	The datasets used for training and evaluating churn prediction models varied widely in sample size, ranging from as few as 61 users to as many as 37 million users (Median = 25,000, IQR = 79,436 - 4,089).	Dataset sizes ranged from 61 to 37 million users, with a median of 25,000 and an IQR of 4,089 to 79,436.	Rearranges and condenses the information for easier and faster understanding.
Technical terms are used without explanation, limiting accessibility.	The findings demonstrate a wide variety of applied ML algorithms - from logistic regression to custom deep neural networks - to predict users' churn probabilities and survival times in mobile apps.	The results show many ML algorithms used for predicting user churn and survival times, from logistic regression to deep neural networks.	Reduces jargon and clarifies the range of algorithms, making the content more accessible.

W4 - Terminology Consistency

Score: 3/5

Terminology is generally consistent, but there are notable lapses, such as inconsistent use and definition of acronyms, variable formatting of technical terms, and ambiguity in key definitions (e.g., 'user churn'). Standardizing terminology, defining acronyms and abbreviations at first use, and ensuring consistent formatting of variables and metrics will improve clarity and professionalism. The manuscript's strength lies in its comprehensive coverage of technical terms, but greater attention to consistency is needed for optimal scholarly communication.

Remarks	Original	Improved	Explanation
The acronym 'ML' is used without definition at first mention, which may confuse some readers.	The acronym 'ML' is used throughout the paper without prior definition.	Machine Learning (ML) is used throughout the paper; ensure to define 'ML' at first mention.	Defining acronyms at first use improves clarity and ensures all readers understand the terminology.
Terms such as 'user churn', 'attrition', 'loss', 'abandonment', and 'dropout' are used interchangeably, causing ambiguity.	Terms such as 'user churn', 'attrition', 'loss', 'abandonment', and 'dropout' are used interchangeably.	Use 'user churn' consistently throughout the manuscript, and define it explicitly as the discontinuation of app usage within the context of this review.	Standardizing terminology and providing a clear definition reduces ambiguity and aids reader understanding.
Performance metrics are reported with inconsistent units and formats, hindering comparison.	Performance metrics like accuracy, F1-Score, and AUC are reported inconsistently.	Standardize reporting as percentages (e.g., Accuracy = 86%) or decimal fractions (e.g., AUC = 0.83), and specify units clearly in the text and tables.	Consistent reporting of metrics facilitates comparison and improves clarity.

W5 - Inclusive Language

Score: **3/5**

The manuscript generally uses neutral and inclusive language, referring to 'users' and 'participants' appropriately. However, there are missed opportunities to address diversity, accessibility, and global representation—such as not considering disability, socioeconomic, or geographic diversity in the analysis and language. Occasional use of labels like 'churners' could be replaced with person-first language. More explicit acknowledgment of inclusivity and diversity would enhance the manuscript's fairness and relevance for a global audience.

Remarks	Original	Improved	Explanation
Labels such as 'churners' or 'disengaged users' can be depersonalizing.	The document predominantly uses 'users' and 'participants' with person-first language, but occasionally uses 'churners' or 'disengaged users.'	Consistently employing person-first language, such as 'users who disengage' or 'participants experiencing churn,' promotes respectful and inclusive terminology.	Person-first language emphasizes individuals over labels, fostering inclusivity.
The review does not explicitly address accessibility or disability considerations.	The review discusses features like 'device type' and 'health features' but does not explicitly consider accessibility or disabilities of users.	While discussing features like 'device type' and 'health features,' future research should explicitly consider accessibility and accommodations for users with disabilities to promote inclusivity.	Encourages the inclusion of accessibility considerations, supporting equitable research.
The search strategy's focus on English-language articles limits geographic inclusivity.	The search strategy focuses on English-language articles, potentially excluding research from non-English speaking regions.	The current search strategy emphasizes English-language publications, which may limit geographic diversity; future efforts should include multilingual sources to broaden global representation.	Acknowledges language bias and promotes global inclusivity.

W6 - Citation Formatting

Score: 3/5

Citation formatting is generally acceptable but marred by inconsistent use of brackets and parentheses, variable placement of citations, and occasional mismatches between in-text citations and the reference list. Standardizing the use of parentheses for all in-text citations, ensuring consistent placement, and thoroughly cross-checking references will enhance the manuscript's professionalism and credibility. The reference list is comprehensive, but uniform formatting is needed for optimal scholarly presentation.

Remarks	Original	Improved	Explanation
Inconsistent use of brackets and parentheses for in-text citations reduces professionalism.	[1, 23, 28, 68, 84]	(1, 23, 28, 68, 84)	Using parentheses uniformly for in-text citations aligns with common academic styles and improves clarity.
Some in-text citations do not clearly match the reference list, hampering source verification.	[12, 77, 80]	(12, 77, 80)	Standardizing citation delimiters and ensuring all citations match the reference list enhances credibility and traceability.
Variable formatting of citation numbers and placement within sentences reduces readability.	Most studies defining churn after a specific cutoff date [1, 23, 28, 68, 84]	Most studies defining churn after a specific cutoff date (1, 23, 28, 68, 84)	Consistent citation formatting improves readability and professional appearance.

W7 - Target Audience Alignment

Score: 4/5

The manuscript is well-aligned with its intended academic audience, offering technical depth, formal language, and comprehensive coverage of methodologies and findings. The structure and terminology are appropriate for researchers and advanced practitioners, though some sections could be made more accessible with clearer subheadings, summaries, and visual aids. Explicitly linking findings to practical implications and including more visual elements would further enhance engagement for both academic and industry readers. Overall, the manuscript's strengths are its rigor and depth, with minor improvements needed for broader accessibility.

Remarks	Original	Improved	Explanation
Dense methodology sections may challenge practitioners seeking procedural clarity.	While detailed, the methodology section is dense and could benefit from clearer subheadings or bullet points to improve readability for practitioners and researchers seeking specific procedural insights.	Add subheadings or bullet points to the methodology section to clearly delineate each step, making it easier for practitioners and researchers to follow and replicate the review process.	Improves navigability and supports practical application by clearly organizing procedural details.
Extensive listing of features and rankings in results may overwhelm non-expert readers.	The extensive listing of features and their importance rankings, while thorough, may overwhelm readers unfamiliar with the domain-specific features, potentially hindering quick comprehension.	Summarize key features and their importance rankings in a table or visual, and provide a brief narrative overview to aid comprehension for readers less familiar with the domain.	Visual summaries and concise overviews improve accessibility for interdisciplinary and practitioner audiences.
Discussion could more explicitly link findings to practical implications for industry stakeholders.	The discussion, though comprehensive, occasionally lacks explicit linkages between findings and practical implications, which could diminish relevance for industry practitioners.	Strengthen the discussion by explicitly connecting research findings to practical implications and recommendations for industry stakeholders.	Enhances the manuscript's utility for applied audiences by making actionable insights more apparent.